The National Higher School of Artificial Intelligence (ENSIA)

Natural Languag Processing

Course Description:

Natural Language Processing addresses fundamental questions at the intersection of human languages and computer science. How can computers acquire, comprehend and produce language? How can computational methods give us insight into observed human language phenomena? In this introductory course, you will learn how computers can do useful things with human languages, such as translate from French into English, filter junk email, extract social networks from the web, and find the main topics in the day's news. You will also learn about how computational methods can help linguists explain language phenomena, including automatic discovery of different word senses and phrase structure. Over the past decade, natural language processing has been revolutionised by statistical and probabilistic methods; you will learn about robust approaches to parameter estimation and inference.

Prerequisite: Machine learning, data mining, advanced programming
Evaluation Method: Coursework (40 %) + Final Exam (60%)

Course Content

Part 1: NLP Fundamentals

- Introduction and Overview
- Language and Text Processing
- N-gram Language Models
- Vector Semantics and Embeddings
- Neural Networks and Neural Language Models
- Sequence Labelling for Parts of Speech and Named Entities
- Transformers and Pre-trained Language Models

Part 2: Annotating Language Structure

- Constituency Parsing
- Dependency Parsing (If time permits)
- Logical Representations of Sentence Meaning (If time permits)
- Computational Semantics and Semantic Parsing (If time permits)

Part 3: NLP Applications (Project-based Learning)

- Machine Translation
- Question Answering
- Chatbots

Part 4: Generative AI for NLP (Project-based Learning)

References

- Dan Jurafsky and James H. Martin, Speech and Language Processing (3rd ed. draft), 2023
- Jacob Eisenstein, "Introduction to Natural Language Processing", The MIT Press, 2019
- Chris Manning and Hinrich Schutze, "Foundations of Statistical Natural Language Processing", MIT Press, 1999

Academic Year 2026–2027
Semester 1
NLP Course Syllabus With Tentative Dates

	Start of S1: 04/10/2026	
Chapter 0 **Course Organisation** + **Chapter 1** **Introduction**	Lectures 1 and 2 04-05/10 **Saturday 10/10**	• Knowledge in Speech and Language Processing • Ambiguity • Models and Algorithms • Language, Thought, and Understanding • The State of the Art • Some Brief History (Individual reading) **MINI PROJECT out**
Chapter 2 **Regular Expressions, Tokenization, Edit Distance**	Lectures 3-4-5 11-12-18/10	• Regular Expressions • Words • Corpora • Simple Unix Tools for Word Tokenization • Word and Subword Tokenization • Word Normalization, Lemmatization and Stemming • Sentence Segmentation • Minimum Edit Distance
Chapter 3 **N-gram Language Models**	Lectures 6–7 19-25/10	• N-Grams • Evaluating Language Models: Training and Test Sets • Evaluating Language Models: Perplexity • Sampling sentences from a language model • Generalizing vs. overfitting the training set • Smoothing, Interpolation, and Backoff • Advanced: Perplexity's Relation to Entropy
Chapter 4 **Naive Bayes, Text Classification, and Sentiment**	Lectures 8-9 26/10 & 2/11	• Naive Bayes Classifiers • Training the Naive Bayes Classifier • Worked example • Optimizing for Sentiment Analysis • Naive Bayes for other text classification tasks • Naive Bayes as a Language Model • Evaluation: Precision, Recall, F-measure • Test sets and Cross-validation • Statistical Significance Testing • Avoiding Harms in Classification
		• The sigmoid function • Classification with Logistic Regression

Chapter 5 **Logistic Regression**	Lecture 10 08/11	• Multinomial logistic regression • Learning in Logistic Regression • The cross-entropy loss function • Gradient Descent • Regularization • Learning in Multinomial Logistic Regression • Interpreting models • Advanced: Deriving the Gradient Equation
Midterm Exams Week 14-21/11/2026		
Chapter 6 **Vector Semantics and Embeddings**	Lectures 11-12-13 09-22-23/11	• Lexical Semantics • Vector Semantics • Words and Vectors • Cosine for measuring similarity • TF-IDF: Weighing terms in the vector • Pointwise Mutual Information (PMI) • Applications of the tf-idf or PPMI vector models • Word2vec • Visualizing Embeddings • Semantic properties of embeddings • Bias and Embeddings • Evaluating Vector Models
Chapter 7 **Neural Networks**	Lecture 14 29/11	• Units • The XOR problem • Feedforward Neural Networks • Feedforward networks for NLP: Classification • Training Neural Nets • Feedforward Neural Language Modeling • Training the neural language model
Chapter 8 **RNNs and LSTMs**	Lecture 15 30/11	• Recurrent Neural Networks • RNNs as Language Models • RNNs for other NLP tasks • Stacked and Bidirectional RNN architectures • The LSTM • Summary: Common RNN NLP Architectures • The Encoder-Decoder Model with RNNs • Attention
Chapter 9 **The Transformer**	Lectures 16-17 06-07/12	• Attention • Transformer Blocks • Parallelizing computation using a single matrix X • The input: embeddings for token and position • The Language Modeling Head

Chapter 10 **Large Language Models**	Lectures 18-19 13–14/12	• Large Language Models with Transformers • Sampling for LLM Generation • Pretraining Large Language Models • Evaluating Large Language Models • Dealing with Scale
Winter Break: from 17/12/2026 to 03/01/2027		
Chapter 11 **Masked Language Models**	Lecture 20 03/01 (26 slides) **Saturday 02/01/2027**	• Bidirectional Transformer Encoders • Training Bidirectional Encoders • Contextual Embeddings • Fine-Tuning for Classification • Fine-Tuning for Sequence Labelling: Named Entity Recognition **Mini project due**
Chapter 12 **Model Alignment, Prompting, and In-Context Learning**	Content to be covered in lab work	• Prompting • Post-training and Model Alignment • Model Alignment: Instruction Tuning • Chain-of-Thought Prompting • Automatic Prompt Optimization • Evaluating Prompted Language Models • Model Alignment with Human Preferences: RLHF and DPO
Chapter 14 **Question Answering, Information Retrieval, and RAG**	Lectures 21-22 04/01	• Information Retrieval • Information Retrieval with Dense Vectors • Answering Questions with RAG • Evaluating Question Answering
Chapter 15 **Chatbots & Dialogue Systems**	Lectures 23-24 10–11/01	• Properties of Human Conversation • Frame-Based Dialogue Systems • Dialogue Acts and Dialogue State • Chatbots • Dialogue System Design
Tuesday **12/01/2027:** No class **Yennayer**		
Chapter 17 **Model Sequence Labeling for Parts of Speech**	Lecture 25 17/01/2027	• (Mostly) English Word Classes • Part-of-Speech Tagging • Named Entities and Named Entity Tagging • HMM Part-of-Speech Tagging • Conditional Random Fields (CRFs)

and Named Entities		• Evaluation of Named Entity Recognition • Further Details
Chapter 18 + 19 **Context-Free Grammars and Constituency Parsing** + **Dependency Parsing** (Dependency Parsing to be covered only if time allows it)	Lectures 23-24 18/01	• Constituency • Context-Free Grammars • Treebanks • Grammar Equivalence and Normal Form • Ambiguity • CKY Parsing: A Dynamic Programming Approach • Span-Based Neural Constituency Parsing • Evaluating Parsers • Heads and Head-Finding + • Dependency Relations • Transition-Based Dependency Parsing • Graph-Based Dependency Parsing • Evaluation
Last day of classes 21/01/2027		
Final Exams Period: 23-30/01/2027		